



CROSS SECTIONAL RETURN PREDICTION

Alpha Tracker

Ankit Kumar

Avanthika Rajesh

Aarush Tapadia

Pranjal Singh





PROJECT OVERVIEW



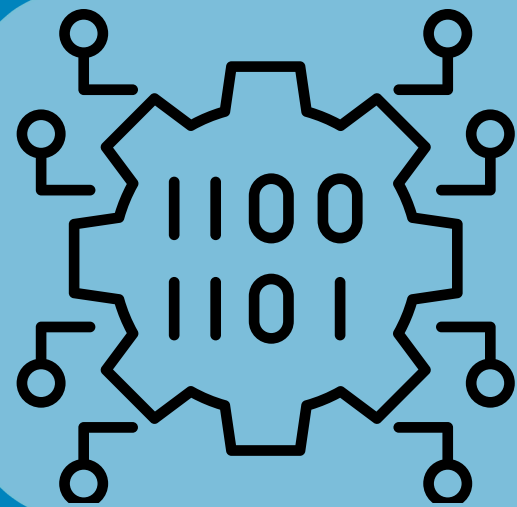
Data Processing & EDA

Handled missing fundamentals via forward-fills and winsorized fat tails to clean the panel.



Feature Engineering

Constructed momentum factors and fundamental ratios, applying cross-sectional Z-scores to prevent look-ahead bias.



Model Selection

Utilized strict time-series cross-validation to evaluate algorithms, ultimately selecting Lasso.



Backtesting & Strategy

Built a dollar-neutral long-short portfolio to evaluate the final Net Sharpe, Rank IC, and turnover.

Project
Overview

Data
Pipeline

EDA & Data
Cleaning

Feature
Engineering

Model
Framework

Validation &
Backtesting

Portfolio
Metrics & risk



DATA PIPELINE

1

2

3

4

Data Sources

- Raw monthly equity panel tracking a universe of 200 stocks.
- Contains roughly 12 years of history.
- Includes split/dividend-adjusted prices, volume, and raw balance-sheet fundamentals.

Data Cleaning

- Identified and dropped duplicate date-ticker rows to ensure panel integrity.
- Forward-filled fundamentals (up to 4 months) to correctly align quarterly reporting with monthly price movements.
- Winsorized features at the 1st and 99th percentiles to mitigate the impact of heavy tails and outliers.

Feature Engineering

- Momentum: Calculated trailing 1m, 3m, 6m, 12m, and 12-1 month price returns.
- Fundamentals: Derived key ratios including P/B, P/S, Debt/Equity, ROA, Operating Margin, and Earnings Yield.
- Normalization: Applied cross-sectional Z-scoring within each month to standardize features against peers and prevent data leakage.

Train Test Split

- Chronological Split: Preserves temporal order to eliminate look-ahead bias.
- Train Set: Months 1–120 (~24k rows); labeled with 1-month forward returns.
- Test Set: Months 121–144 (~4.8k rows); withheld for final scoring.
- Validation: 5-fold expanding window time-series split on training data

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Pipeline

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Feature
Engineering

Model
Framework

Validation &
Backtesting

Portfolio
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EDA AND DATA CLEANING

Missing Values

- Finding: ~6.8%–7.1% missingness in fundamental fields due to quarterly reporting lag.
- Action: Forward-filled fundamentals up to 4 months (`ffill(limit=4)`) per ticker; remaining NaNs imputed to 0 post-normalization.

Duplicate Values

- Identified 80 duplicate (date, ticker) rows in train.csv that corrupted panel balance.
- Action: Cleaned using `.drop_duplicates(subset=["date", "ticker"], keep="last")` to maintain a strict 200-stock monthly cross-section.

Feature Engineering

- Momentum: Trailing 1m, 3m, 6m, 12m, and 12-1 month returns.
- Fundamentals: Value (P/B, P/S, Earnings Yield), Quality (ROA, Operating Margin), Leverage (Debt/Assets, Debt/Equity), and Size (Log Market Cap).
- Flags: Added explicit binary missingness flags for imputed fields.

Outliers and Fat tails

- Heavy tails in raw ratios (e.g. P/B max = 37.3, Debt/Equity max = 19.9) distorted raw signals.
- Action: Applied monthly Winsorization at the 1st and 99th percentiles (`clip(p01, p99)`) to cap extreme outliers

Cross Sectional Alignment

- Raw financial metrics span orders of magnitude and carry non-stationary market regime shifts.
- Action: Standardized all features within each month via cross-sectional Z-score ($\mu=0$, $\sigma=1$) across the 200 stocks to isolate peer rankings.

Time Series Alignment

- Computing 12-month momentum for early test dates requires look-back into the end of the training set.
- Concatenated train and test panels, sorted strictly by (ticker, date), and applied rolling window calculations to ensure zero look-ahead leakage.

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Pipeline

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Model
Framework

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Portfolio
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FEATURE ENGINEERING

Value

- Derived Price-to-Book and Price-to-Sales to capture relative cheapness.
- Calculated Earnings Yield by dividing trailing EPS by the adjusted close price.

Momentum

- Computed trailing price returns across 1m, 3m, 6m, and 12m horizons.
- Isolated the 12-1 month momentum to benchmark against standard trend-following signals.

Quality

- Measured operational efficiency using Return on Assets (ROA).
- Evaluated core profitability by generating the Operating Margin.

Leverage & Size

- Assessed balance sheet risk via Debt-to-Assets and Debt-to-Equity ratios.
- Extracted the Log Market Cap to account for cross-sectional size premiums.

Project
Overview

Data
Pipeline

EDA & Data
Cleaning

Feature
Engineering

Model
Framework

Validation &
Backtesting

Portfolio
Metrics & risk



MODEL FRAMEWORK

Model Selection and Rationale

- **Evaluated Models:** Tested Linear Regression, Ridge, Lasso, Elastic Net, Random Forest, and Gradient Boosting.
- **Final Selection:** Lasso Regression.
- **Rationale:** Lasso achieved the highest Net Sharpe (1.818) and ICIR (0.968). Tree-based models (Random Forest, GBM) were rejected due to excessive turnover and severe drawdowns.

Target Variable Formulation

- **Target Label:** fwd_ret_1m, which represents the realized return over the following month.
- **Objective:** Output a single real-valued score per stock to predict the relative cross-sectional ordering of peers.
- **Scope:** Focused exclusively on ranking which names will outperform, ignoring absolute price or broad market direction.

Hyperparameter Tuning

- **Validation Strategy:** Implemented a 5-fold expanding window time-series cross-validation.
- **Data Integrity:** Strictly enforced chronological splits (no random shuffling) to completely eliminate look-ahead bias.
- **Parameters:** The Lasso regularization penalty (alpha) was set to 0.001 to optimize the bias-variance tradeoff.

Feature Importance and Selection

- **Automated Selection:** Lasso's L1 regularization inherently performs feature selection by shrinking uninformative coefficients entirely to zero.
- **Feature Set:** Combined momentum signals (e.g., mom_12_1) with fundamental financial ratios.
- **Preparation:** Relied on cross-sectionally Z-scored features to isolate peer outperformance and handle differing metric scales.

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Overview

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Pipeline

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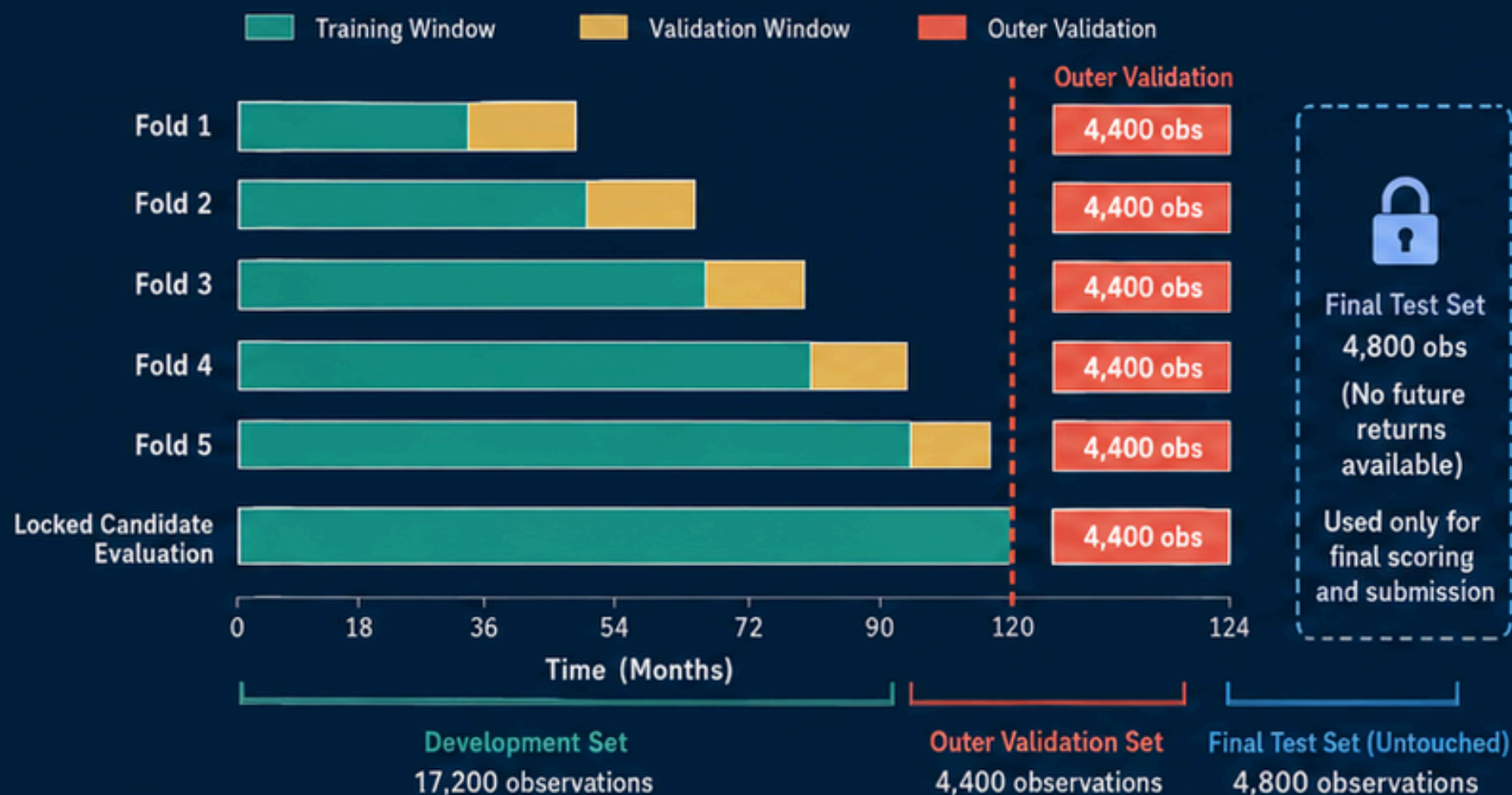
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Portfolio
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VALIDATION AND BACKTESTING

Time-Series Cross-Validation



5-fold expanding-window CV was performed within the development set for model and factor selection. The locked candidate was then evaluated once on the untouched outer-validation set.



Historical Backtest Simulation



Factor A = 50% Momentum + 20% Value + 10% Quality + 20% Leverage | Lasso $\alpha = 0.001$

OUTER VALIDATION PERFORMANCE



Final Candidate: Factor A + 10% Lasso

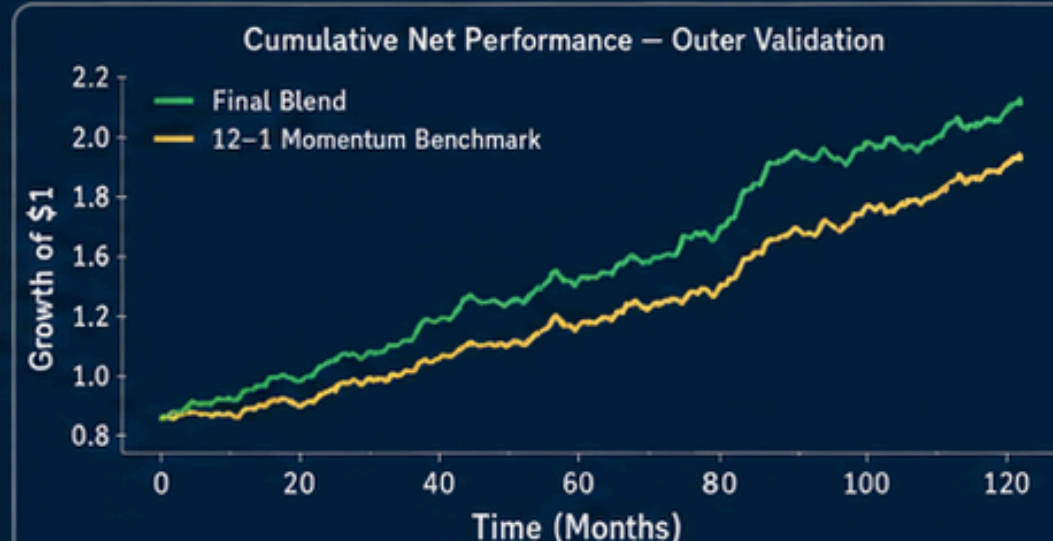
BENCHMARK COMPARISON

FINAL BLEND
1.818
Net Sharpe

VS

12-1 MOMENTUM
1.780
Net Sharpe

Final blend outperformed the official 12-1 momentum benchmark on outer-validation Net Sharpe.



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PORTFOLIO METRICS AND RISK

Information Coefficient Analysis



Measures how well model scores predict next-month realized returns.

RANK IC

0.0633

Monthly Spearman correlation

ICIR

0.968

Mean IC ÷ Std Dev of IC



High ICIR indicates consistent predictive skill across months, rather than performance being driven by a few lucky periods.

FINAL CANDIDATE

Factor A + 10% Lasso

Lasso $\alpha = 0.001$

Turnover and Cost Penalisation

MEAN TURNOVER (LOWER IS BETTER)

FINAL BLEND

90% Factor A + 10% Lasso

MEAN TURNOVER
1.287x



10 bps transaction cost per unit of turnover



Turnover is calculated from month-over-month changes in portfolio weights and transaction costs are applied at 10 bps per unit of turnover.

1.287x is the final mean turnover.

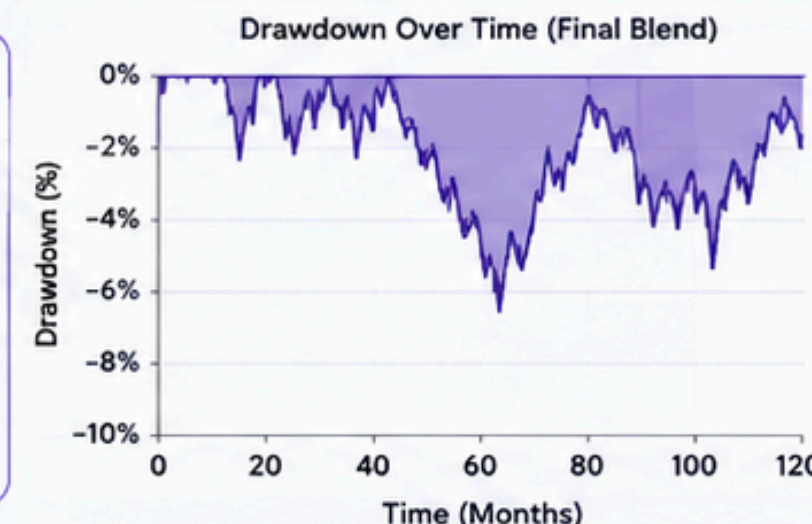
Drawdown and Tail Risk

MAX DRAWDOWN (LOWER IS BETTER)

FINAL BLEND

90% Factor A + 10% Lasso

MAX DRAWDOWN
-5.48%



Final blend maintains strong predictive performance while keeping downside risk controlled.

FINAL OUTER-VALIDATION RESULTS

NET SHARPE

1.818

RANK IC

0.0633

ICIR

0.968

MEAN TURNOVER

1.287x

MEAN MONTHLY NET RETURN

1.44%

MAX DRAWDOWN

-5.48%

FINAL CANDIDATE

Factor A + 10% Lasso

Lasso $\alpha = 0.001$

BENCHMARK COMPARISON (OUTER VALIDATION)

FINAL BLEND
(FACTOR A + 10% LASSO)

NET SHARPE

1.818

VS

OFFICIAL 12-1 MOMENTUM
BENCHMARK

NET SHARPE

1.780

1.818 > 1.780



Final blend outperformed the official 12-1 momentum benchmark on outer-validation Net Sharpe.

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Framework

Validation &
Backtesting

Portfolio
Metrics & risk



THANK YOU